

C1.4 What choices can we make personally, locally, nationally or globally to improve air quality?

1. understand how atmospheric pollution caused by power stations which burn fossil fuels can be reduced by:
 - using less electricity;
 - removing sulfur from natural gas and fuel oil;
 - removing sulfur dioxide and particulates (carbon and ash) from the flue gases emitted by coal-burning power stations;
2. understand that the only way of producing less carbon dioxide is to burn less fossil fuels;
3. understand how atmospheric pollution caused by exhaust emissions from motor vehicles can be reduced by:
 - burning less fuel by having more efficient engines;
 - using low sulfur fuels;
 - using catalytic converters, which convert nitrogen monoxide to nitrogen and oxygen and carbon monoxide to carbon dioxide;
 - adjusting the balance between public and private transport;
 - having legal limits to emissions (which are enforced by the use of MOT tests);
4. in the context of emissions of pollutants into the atmosphere:

Our way of life depends on a wide range of materials produced from natural resources. This module considers how measurements of the properties of materials can inform the choice of material for a particular purpose. By taking their own measurements, candidates explore some of the issues which arise when trying to establish accurate and meaningful data.

Key ideas in this module are illustrated through polymers. Candidates learn how the particles (e.g. molecules) that make up a material fit together and how strongly they hang on to each other, providing an explanation of the properties of materials. This provides an example of a scientific explanation which makes sense of a wide range of observations.

Through conducting a life cycle assessment, candidates learn that in selecting a product for a particular job we should assess not only its 'fitness for purpose' but also the total effects of using the materials that make up the product over its complete life cycle, from its production from raw materials to its disposal.

Issues for citizens	Questions that science may help to answer
How can we pick a suitable material for a particular product or task?	What different properties do different materials have? Why is crude oil important as a source of new materials such as plastics and fibres? Why does it help to know about the molecular structure of materials such as plastics and fibres?
When buying a product, what else should we consider besides its cost and how well it does its job?	How should we manage the wastes that arise from our use of materials?
Science Explanations	Ideas about Science
SE 3 Materials and their properties	IaS 1 Data and its limitations IaS 2.2 Correlation and cause IaS 6.1–6.4, 6.7 Making decisions about science and technology

ICT Opportunities

This module offers opportunities for illustrating the use of ICT in science. For example, logging data, storing it and displaying it in a variety of formats for analysis and evaluation

Use of ICT in teaching and learning can include:

- spreadsheet to record and display measurements of the properties of materials;
- video clips to illustrate main stages from extraction of oil to production of synthetic plastic or fibre;
- still images and diagrams to create presentations to show how the properties of a material depend on its molecular structure;